

Singular direction (**Singular.lean**)

$\inf_s \prod_{i \in s} H_i = 0$: extract finite windows s_k
with partial affinity product $a_k < 4^{-k}$

cylinder events $B_k = \{Z_{s_k} < 4^{-k}\}$,
 $Z_s(x) = \prod_{i \in s} (d\mu_i/d\nu_i)(x_i)$

$\mu^\infty(B_k) \leq 2^{-k}$ (cylinder density identity)
 $\nu^\infty(B_k^c) \leq 2^{-k}$ (Markov on $\sqrt{Z_{s_k}}$ + moment identity)

Borel–Cantelli, twice:
 $S = \limsup B_k$ separates, so $\mu^\infty \perp_m \nu^\infty$

Absolutely continuous direction (**Equivalence.lean**)

summable deficits: **Summable.vanishing** picks a finite
window t with $\sum_{i \notin t} (1 - H_i) < \eta$ (the one $\mathbb{R}_{\geq 0}^\infty \rightarrow \mathbb{R}$ crossing)

second moment $\int Z_t(\sqrt{W} - 1)^2 d\nu^\infty \leq 2(1 - \prod H_i)$
+ Cauchy–Schwarz, entirely in $\mathbb{R}_{\geq 0}^\infty$

quantitative cylinder comparison: on every cylinder,
 $\mu^\infty(A) \leq (\nu^\infty.\text{withDensity } Z_t)(A) + 2(2 - 2\prod H_i)^{1/2}$

extend over the measure-dense cylinder algebra;
let the tail vanish: $\nu^\infty(E) = 0 \implies \mu^\infty(E) = 0$